

Yashveer Jain

[yashveerjain.github.io](https://github.com/yashveerjain) | [linkedin.com/in/yashveerjain/](https://www.linkedin.com/in/yashveerjain/) | yashveer@terpmail.umd.edu

EDUCATION

Master of Engineering (Robotics) (GPA:3.87/4) University of Maryland – College Park (UMD), USA	Aug 2022–May 2024
Bachelor of Engineering (Mechatronics) (GPA:9.44/10) Manipal University Jaipur, Rajasthan, India Courses: Industrial Robotics, Pneumatics and Hydraulics, MEMS, PLC, Analog and Digital Circuit.	July 2016- July 2020

WORK EXPERIENCE

Embedded AI Intern - Renesas Electronics, USA - Trained and deployed SVM and logistic regression models on Renesas RA6 Series and RL78 microcontroller, achieving 95% accuracy for an Asset Tracking Project, enabling efficient asset management. - Ensured success of edge computing projects by developing and implementing machine learning models optimized for resource-constrained embedded systems.	May 2023–Aug 2023
Machine learning Engineer - AIMonk Labs, India - Optimized OCR algorithms using Transformers implemented on PyTorch, achieving a 60% accuracy boost, significantly improving text recognition capabilities. - Enhanced car detection AI by 20% using DeepLabv3 and YOLOv7 integration, leading to more accurate object detection in autonomous systems. - Collaborated with cross-functional teams on Azure DevOps for model deployment on AWS SageMaker, ensuring seamless integration and deployment of machine learning models.	Aug 2020–Aug 2022
Research Intern - Swaayatt Robotics, India - Implemented Kalman and particle filters with LK optical flow using OpenCV for vehicle and pedestrian tracking in autonomous driving systems. - Implemented UnDeepVO paper unsupervised deep learning visual odometry paper using TensorFlow 2.0 and Keras for depth estimation and translation and rotation motion, increasing the SLAM accuracy by 20%.	Jan 2020–May 2020

RESEARCH AND TEACHING EXPERIENCE

Graduate Research Assistant, University of Maryland (Spring 2024) AAM-SEALS: Developing Aerial-Aquatic Manipulators in Sea, Air, and Lands Simulator. Under Review CoRL. A photorealistic simulator based on Isaac Sim for Aerial-Aquatic Mobile Manipulators (AAMs) in sea, air, and land environments, enhancing integrated controllers for drone and hydrodynamics simulations.	Jan 2024-May 2024
Graduate Teaching Assistant, University of Maryland (Fall 2023) ENPM667 – Control of Robotic Systems with Prof. Waseem Malik, a graduate course at UMD. Providing targeted tutoring in Linearization techniques and control system topics like PID, LQR, LQG.	Aug 2023-Dec 2023
Graduate Research Assistant, University of Maryland (Fall 2022-Spring 2023) TinyDepth: Generalized Neural Metric Depth for Palm-sized Robots. Under Review Nature npj Robotics. Designed sub-6cm microdrone with embedded Raspberry Pi and ROS-based vision pipeline for autonomous planning and control (Perception & Robotics Group, Prof. Yiannis Aloimonos).	Aug 2022-May 2023

PROJECTS

SVM-based Image Classification with Multithreading (C++)- GitHub Implemented a Support Vector Machine (SVM) for image classification in C++. Utilized multithreading for improved performance on the CIFAR-10 dataset.	Mar 2024-May 2024
ARIAC Competition (conducted by NIST) - GitHub Developed robot control algorithm (C++, ROS2, MoveIt!) for ARIAC Kitting Task. Implemented YOLO for object detection (part & tray) And ensured seamless robot-software communication (ROS2 backend).	Aug 2022-May 2023
Robo-Serve - GitHub Designed and deployed autonomous delivery robot using Solidworks, ROS, differential drive controller, and A* pathfinding. Project improved hygiene and safety in food service during COVID-19. Skills: Solidworks, ROS, navigation algorithms, robot kinematics.	Aug 2022-May 2023
Home Automation - GitHub Established a home server for device connectivity using FastAPI, MQTT, Arduino, Raspberry Pi. Demonstrating proficiency in hardware integration and IoT systems.	Jun 2021-Aug 2021

SKILLS AND LANGUAGES

Machine Learning: TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV, Scikit-learn, Vector-c++, Huggingface.
Programming Languages: Python, C, C++, Ladder Programming
Development Tools: Git, Docker, Agile Development, AWS (Certified Cloud Practitioner), Azure.
Robotics: ROS (Robot Operating System), ROS2, Arduino, Raspberry Pi, Nvidia Jetson, Robot Modelling, LQR, PID Control System, Motion Planning, Graph Theory, SLAM (particle filter/ Kalman filter and occupancy grid mapping), CAD Tools (Solidworks, Fusion360).